
DSC 291 Trustworthy ML

Course Proposal Template (Research Reproduction)

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1 Introduction

Project format: Tutorial on Research Reproduction (Format 1).

Large language models (LLMs) are aligned to refuse harmful requests, but this alignment can be bypassed by *jailbreaks*—adversarial prompts that elicit objectionable content. Existing jailbreak attacks fall into two categories, both with significant drawbacks: prompt-level attacks rely on hand-crafted social-engineering prompts that require extensive human effort, while token-level attacks (e.g., GCG Zou et al. [2023]) optimize gibberish suffixes that need white-box access and hundreds of thousands of queries to the target model. Identifying jailbreak vulnerabilities efficiently is essential for stress-testing alignment before LLMs are deployed in safety-critical settings.

We propose to reproduce **Prompt Automatic Iterative Refinement (PAIR)** Chao et al. [2023], a black-box jailbreak attack that bridges these two categories. PAIR pits an *attacker* LLM against a *target* LLM: the attacker generates a candidate adversarial prompt, observes the target’s response, and iteratively refines the prompt using chain-of-thought reasoning. The paper’s main contributions are: (i) PAIR is query-efficient, typically finding a successful jailbreak in fewer than 20 queries—orders of magnitude fewer than GCG; (ii) it is effective across both open- and closed-source models, achieving 88% attack success rate (ASR) on Vicuna-13B, 51% on GPT-3.5, 48% on GPT-4, and 73% on Gemini-Pro on the JailbreakBench behaviors; and (iii) the resulting prompts are human-interpretable and transfer well across models. Because PAIR runs entirely through black-box API queries (the authors report $\sim \$0.03$ and 366 MB of CPU memory per jailbreak when attacking Vicuna-13B), it is a particularly tractable target for a class reproduction.

2 Proposal

We will use the official PAIR codebase¹ with the paper’s v4 (JailbreakBench) configuration: Mixtral-8x7B-Instruct as attacker, $N=30$ streams of depth $K=3$ (max 90 queries per behavior), attacker temperature 1.0 / top- $p=0.9$, target temperature 0, $L=150$ generated tokens, and Llama Guard as JUDGE (with Llama-3-70B as a sensitivity check because the JailbreakBench repository has since switched its default classifier); evaluation uses the 100-behavior JBB-Behaviors set. Our reproduction covers the paper’s three claimed contributions—*effectiveness*, *efficiency*, and *transferability*—with one artifact per claim plus a category-level breakdown. We deliberately exclude the ablations (Tables 6–7, Figs. 5–6) and the defense experiment (Table 5), since the latter introduces extra systems (SmoothLLM and a perplexity filter) whose own reproduction error would obscure the PAIR signal. We similarly exclude Table 1 (judge selection), which depends on a 100-pair human-annotation effort outside the scope of an algorithm reproduction.

¹<https://github.com/patrickrchao/JailbreakingLLMs>

An immediate reproducibility constraint. A strict 1:1 reproduction of the paper’s seven target models is no longer possible: Claude-instant-1.2 (retired November 2024) and Claude-2.1 (retired July 2025) are no longer callable, the GPT-4 snapshot used in the paper (gpt-4-0125-preview) was shut down on March 26, 2026, and the original Gemini-Pro (Gemini 1.0 Pro) endpoint has been retired in favor of the 1.5/2.x/3.x generations. Rather than paper over this, we treat it as part of the proposal: (i) the main reproduction restricts attention to targets whose weights have not changed (Vicuna-13B-v1.5, Llama-2-7B-chat, GPT-3.5-Turbo, snapshot gpt-3.5-turbo-1106); (ii) the retired models become the motivation for our additional experiment described below.

Reproduction targets.

- **Table 2 — Direct jailbreak attacks (effectiveness).** The headline result. We will report Jailbreak % and Queries-per-Success on Vicuna-13B-v1.5, Llama-2-7B-chat and GPT-3.5-Turbo. The key quantity to verify is the gap between Vicuna (88%) and Llama-2 (4%), which the paper attributes to safety fine-tuning. Reproducing GCG is out of scope due to its ~256,000-query, GPU-hours-per-behavior cost; we will reuse the paper’s reported GCG numbers for context.
- **Table 3 — Transferability.** Using successful jailbreaks generated against Vicuna in the previous step, we will measure transfer Jailbreak % on Llama-2 and GPT-3.5. The paper’s central claim is that PAIR’s semantic prompts transfer better than GCG’s token-level suffixes; we will check whether the paper’s asymmetric pattern replicates qualitatively (e.g., Vicuna→GPT-3.5 transferring at 52% while Vicuna→Llama-2 collapses to 1%).
- **Table 4 — Efficiency.** We will measure wall-clock time, memory footprint, and per-jailbreak API cost on Vicuna-13B and check the paper’s headline ~34 s / 366 MB / \$0.03 figures. Because PAIR’s runtime is dominated by API latency rather than compute, we expect the wall-clock number to be the most variable across reproductions and will report it with confidence intervals over multiple runs.
- **Figure 4 — Per-category Jailbreak %.** A heatmap of Jailbreak % across the 10 JBB-Behaviors misuse categories and the three reproduction target models. This is the most informative artifact for the tutorial: it shows *where* alignment is weakest (e.g., Malware/Hacking, Government Decision-Making) and is also visually compelling for the Medium write-up.

Additional experiment: longitudinal robustness. The retirement or replacement of four of seven target models means the reported numbers can no longer be verified against the original endpoints. We turn this into our main extension. Using PAIR *unmodified*, we will run it against the modern successors of the retired and replaced closed-source targets: GPT-4o (gpt-4o-2024-11-20, replacing gpt-4-0125-preview), Claude Sonnet 4.6 (replacing Claude-1/2), and Gemini 2.5 Pro (replacing Gemini-Pro), on the same 100-behavior set with identical hyperparameters. **Hypothesis:** Jailbreak % on the closed-source successors has dropped substantially relative to the 2024 numbers (consistent with two years of safety post-training), while Queries-per-Success on the jailbreaks that *do* succeed has changed less, because PAIR’s iterative refinement adapts to whatever defense is in place: it just gives up more often on hardened categories. **Reported metrics:** (a) Jailbreak %, (b) Queries-per-Success, and (c) per-category breakdown to identify which misuse categories are now patched versus which remain vulnerable. The experiment reuses the existing pipeline; cost is bounded by API calls (< \$50).

References

- Patrick Chao, Alexander Robey, Edgar Dobriban, Hamed Hassani, George J. Pappas, and Eric Wong. Jailbreaking black box large language models in twenty queries. *arXiv preprint arXiv:2310.08419*, 2023.
- Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.